

Linear Programming Graphical Method



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Operations Research (OR)

- Operations research (OR) is an analytical method of problem-solving and decision-making that is useful in the management of organizations.
- In operations research, problems are broken down into basic components and then solved in defined steps by mathematical analysis.
- The concept of operations research arose during World War II by military planners.
- After the war, the techniques used in their operations research were applied to addressing problems in business, the government and society.

Operations Research (OR)

- The process of operations research can be broadly broken down into the following steps:
 - Identifying a problem that needs to be solved.
 - Constructing a model around the problem that resembles the real world and variables.
 - Using the model to derive solutions to the problem.
 - Testing each solution on the model and analyzing its success.
 - Implementing the solution to the actual problem.

Operations Research (OR)

- There are three primary characteristics of all operations research efforts:
- **Optimization-** The purpose of operations research is to achieve the best performance under the given circumstances. Optimization also involves comparing and narrowing down potential options.
- **Simulation-** This involves building models or replications in order to try out and test solutions before applying them.
- **Probability and statistics-** This includes using mathematical algorithms and data to uncover helpful insights and risks, make reliable predictions and test possible solutions.

Operations Research (OR)

- Operations research can be applied to a variety of use cases, including:
 - ✓ Scheduling and time management.
 - ✓ Urban and agricultural planning.
 - ✓ Enterprise resource planning (ERP) and supply chain management (SCM).
 - ✓ Inventory management.
 - ✓ Network optimization and engineering.
 - ✓ Packet routing optimization.
 - ✓ Risk management.

Linear Programming

- linear programming, mathematical modeling technique in which a linear function is maximized or minimized when subjected to various constraints.
- This technique has been useful for guiding quantitative decisions in business planning, in industrial engineering, and—to a lesser extent—in the social and physical sciences.
- Linear Programming Problems (LPP) provide the method of finding such an optimized function along with/or the values which would optimize the required function accordingly.
- Formulation of an LPP refers to translating the real-world problem into the form of mathematical equations which could be solved.
- It usually requires a thorough understanding of the problem.

Steps of Formulating Linear Programming

- **Step 1:** Identify the 'n' number of decision variables which govern the behaviour of the objective function (which needs to be optimized).
- **Step 2:** Identify the set of constraints on the decision variables and express them in the form of linear equations /inequations. This will set up our region in the n-dimensional space within which the objective function needs to be optimized. Don't forget to impose the condition of non-negativity on the decision variables i.e. all of them must be positive since the problem might represent a physical scenario, and such variables can't be negative.
- **Step 3:** Express the objective function in the form of a linear equation in the decision variables.
- **Step 4:** Optimize the objective function either graphically or mathematically.

Example – Formulating LP

- A calculator company produces a handheld calculator and a scientific calculator. Long-term projections indicate an expected demand of at least 150 scientific and 100 handheld calculators each day. Because of limitations on production capacity, no more than 250 scientific and 200 handheld calculators can be made daily. To satisfy a shipping contract, a minimum of 250 calculators must be shipped each day. If each scientific calculator sold, results in a 20 rupees loss, but each handheld calculator produces a 50 rupees profit; then how many of each type should be manufactured daily to maximize the net profit?



Example – Formulating LP

- **Step 1:** The decision variables: Since the question has asked for an optimum number of calculators, that's what our decision variables in this problem would be. Let,
 - x = number of scientific calculators produced
 - y = number of handheld calculators produced
- Therefore, we have 2 decision variables in this problem, namely ' x ' and ' y '.

Example – Formulating LP

- **Step 2:** The constraints: Since the company can't produce a negative number of calculators in a day, a natural constraint would be:
 - $x \geq 0$
 - $y \geq 0$
- However, a lower bound for the company to sell calculators is already supplied in the problem. We can note it down as:
 - $x \geq 150$
 - $y \geq 100$

Example – Formulating LP

- We have also been given an upper bound for these variables, owing to the limitations on production by the company. We can write as follows:
 - $x \leq 250$
 - $y \leq 200$
- Besides, we also have a joint constraint on the values of 'x' and 'y' due to the minimum order on a shipping consignment; given as:
 - $x + y \geq 250$

Example – Formulating LP

- **Step 3:** Objective Function: Clearly, here we need to optimize the Net Profit function. The Net Profit Function is given as:

$$P = -20x + 50y$$

Example – Formulating LP

- Step 4: Solving the problem: the system here –

Maximization of $P = -20x + 50y$,

subject to:

$$150 \leq x \leq 250$$

$$100 \leq y \leq 200$$

$$x + y \geq 250$$

Types of Linear Programming Problems

- **Diet Problems**

- As the name suggests in itself, such problems involve optimizing the intake of certain types of foods rich in certain nutrients that could help one follow a particular diet plan.
- More precisely, the goal of a diet problem is to select a set of foods that will satisfy a set of a daily nutritional requirement at a minimum cost.
- Constraints – The specified nutritional requirements, that could be a specific calorie intake or the amount of sugar or cholesterol in the diet.
- Objective function – The cost of the food intake.

Types of Linear Programming Problems

- Example: A kitchen manager at Nulife Hospital has to decide the food mix for the patients. Dietary instructions are that each patient must get at least:
 - One gram of protein
 - One gram of fat
 - Three gms. of carbohydrates

Types of Linear Programming Problems

- **Manufacturing Problems**

- These problems involve optimizing the production rate or the net profits of the manufactured products, which could be a function of the available workspace, the number of labourers, machine hours, packaging material used, raw materials required, the market value of the product etc.
- These find application in the industrial sectors and hence the prediction of a company's possible capital increase over the years.
- Constraints – Variables like the work-hours, the cost of the packaging material used etc.
- Objective function – The production rate

Types of Linear Programming Problems

- Example: A company makes two products (X and Y) using two machines (A and B).
- Each unit of X that is produced requires 50 minutes processing time on machine A and 30 minutes processing time on machine B.
- Each unit of Y that is produced requires 24 minutes processing time on machine A and 33 minutes processing time on machine B.
- At the start of the current week, there are 30 units of X and 90 units of Y in stock. Available processing time on machine 'A' is forecast to be 40 hours and on machine 'B' is forecast to be 35 hours.
- The forecast of demand for X in the current week is 75 units and for Y is 95 units.
- Company policy is to maximize the combined sum of the units of X and the units of Y in stock at the end of the week.

Types of Linear Programming Problems

- **Transportation Problems**

- These problems are related to the study of the efficient transportation routes i.e. how efficiently the product from different sources of production is transported to the different markets, such as the total transportation cost is minimum.
- Analysis of such problems is very crucial for big companies with several production plants and a widespread area to cater to.
- Constraints – The specific supply and demand patterns.
- Objective function – The transportation cost

Types of Linear Programming Problems

- Example: A firm has 3 factories – Alpha, Tango, and Kingston.
- There are four major warehouses situated at locations PTR, CNB, DLS, and MMB.
- The average daily production of the product at Alpha, Tango, Kingston is 30, 40 and 50 units respectively.
- The average daily requirement of this product at PTR, CNB, DLS, and MMB is 35, 30, 32, 25 units respectively.
- The goal here is to minimize the total transportation costs.

Types of Linear Programming Problems

- **Optimal Assignment Problems**

- These problems are concerned with the completion of a particular task/assignment of a company by choosing a certain number of employees to complete the assignment within the required deadline, given that a single person works on only one job within the assignment.
- Such problems find their application in event planning/management in big companies etc.
- Constraints – The number of employees, the work-hours of each employee etc.
- Objective function – The total assignments done

Types of Linear Programming Problems

- Example: The Watson Toys and Watches Company has four men available for work on four separate jobs.
- Only one man can work on any one job.
- Find a suitable system of assigning men to the jobs in such a way that the total cost of the assignment is minimum.

Linear Programming – Graphical Method

- Owing to the importance of linear programming models in various industries, many types of algorithms have been developed over the years to solve them.
- Some famous mentions include the Simplex method, the Hungarian approach, and others.
- Here we are going to concentrate on one of the most basic methods to handle a linear programming problem i.e. the graphical method.
- In principle, this method works for almost all different types of problems but gets more and more difficult to solve when the number of decision variables and the constraints increases.
- Therefore, we'll illustrate it in a simple case i.e. for two variables only.

Steps of Graphical Method

- **Step 1: Formulate the LP (Linear programming) problem**
- We have already understood the mathematical formulation of an LP problem in a previous section.
- Note that this is the most crucial step as all the subsequent steps depend on our analysis here.

Steps of Graphical Method

- **Step 2: Construct a graph and plot the constraint lines**
- The graph must be constructed in 'n' dimensions, where 'n' is the number of decision variables.
- This should give you an idea about the complexity of this step if the number of decision variables increases.
- One must know that one cannot imagine more than 3-dimensions anyway.
- The constraint lines can be constructed by joining the horizontal and vertical intercepts found from each constraint equation.

Steps of Graphical Method

- **Step 3: Determine the valid side of each constraint line**
- This is used to determine the domain of the available space, which can result in a feasible solution.
- A simple method is to put the coordinates of the origin (0,0) in the problem and determine whether the objective function takes on a physical solution or not.
- If yes, then the side of the constraint lines on which the origin lies is the valid side.
- Otherwise it lies on the opposite one.

Steps of Graphical Method

- **Step 4: Identify the feasible solution region**
- The feasible solution region on the graph is the one which is satisfied by all the constraints.
- It could be viewed as the intersection of the valid regions of each constraint line as well.
- Choosing any point in this area would result in a valid solution for our objective function.

Steps of Graphical Method

- **Step 5: Plot the objective function on the graph**
- It will clearly be a straight line since we are dealing with linear equations here.
- One must be sure to draw it differently from the constraint lines to avoid confusion.
- Choose the constant value in the equation of the objective function randomly, just to make it clearly distinguishable.

Steps of Graphical Method

- **Step 6: Find the optimum point**
- An optimum point always lies on one of the corners of the feasible region.
- How to find it? Place a ruler on the graph sheet, parallel to the objective function.
- Be sure to keep the orientation of this ruler fixed in space.
- We only need the direction of the straight line of the objective function.
- Now begin from the far corner of the graph and tend to slide it towards the origin.

Steps of Graphical Method

- **Step 6: Find the optimum point**
 - **If the goal is to minimize the objective function**, find the point of contact of the ruler with the feasible region, which is the closest to the origin.
 - This is the optimum point for minimizing the function.
 - **If the goal is to maximize the objective function**, find the point of contact of the ruler with the feasible region, which is the farthest from the origin.
 - This is the optimum point for maximizing the function.

Steps of Graphical Method

- **Step 7: Calculate the coordinates of the optimum point**
- This is the last step of the process.
- Once you locate the optimum point, you'll need to find its coordinates.
- This can be done by drawing two perpendicular lines from the point onto the coordinate axes and noting down the coordinates.
- Otherwise, you may proceed algebraically also if the optimum point is at the intersection of two constraint lines and find it by solving a set of simultaneous linear equations.
- The Optimum Point gives you the values of the decision variables necessary to optimize the objective function.
- To find out the optimized objective function, one can simply put in the values of these parameters in the equation of the objective function.

Graphical Method Example

- A health-conscious family wants to have a very well controlled vitamin C-rich mixed fruit-breakfast which is a good source of dietary fibre as well; in the form of 5 fruit servings per day. They choose apples and bananas as their target fruits, which can be purchased from an online vendor in bulk at a reasonable price. Bananas cost 30 rupees per dozen (6 servings) and apples cost 80 rupees per kg (8 servings). Given: 1 banana contains 8.8 mg of Vitamin C and 100-125 g of apples i.e. 1 serving contains 5.2 mg of Vitamin C. Every person of the family would like to have at least 20 mg of Vitamin C daily but would like to keep the intake under 60 mg. How much fruit servings would the family have to consume on a daily basis per person to minimize their cost?



Graphical Method Example

- We begin step-wise with the formulation of the problem first. The constraint variables – ‘x’ = number of banana servings taken and ‘y’ = number of servings of apples taken.
- Let us find out the objective function now.
- Cost of a banana serving = 30/6 rupees = 5 rupees. Thus, the cost of ‘x’ banana servings = 5x rupees
- Cost of an apple serving = 80/8 rupees = 10 rupees. Thus the cost of ‘y’ apple servings = 10y rupees

$$\textit{Total Cost } C = 5x + 10y$$

Constraints: $x \geq 0$; $y \geq 0$ (non-negative number of servings)

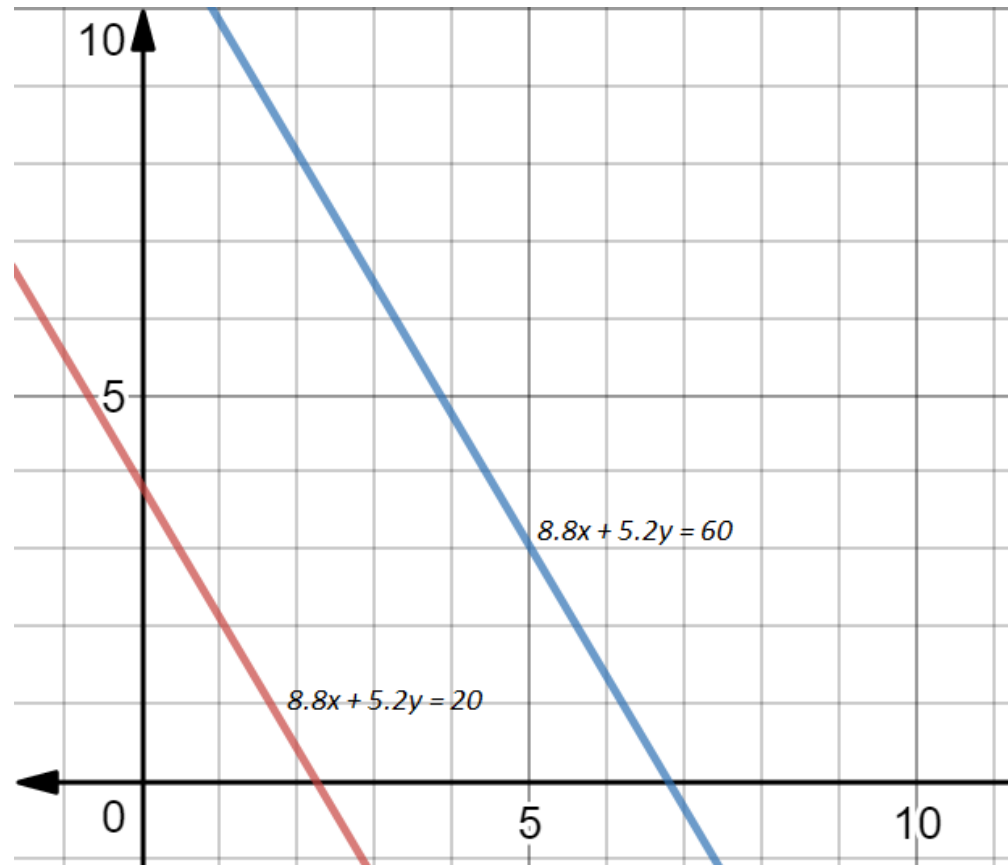
- Total Vitamin C intake:

$$8.8x + 5.2y \geq 20 \quad (1)$$

$$8.8x + 5.2y \leq 60 \quad (2)$$

Graphical Method Example

- Now let us plot a graph with the constraint equations

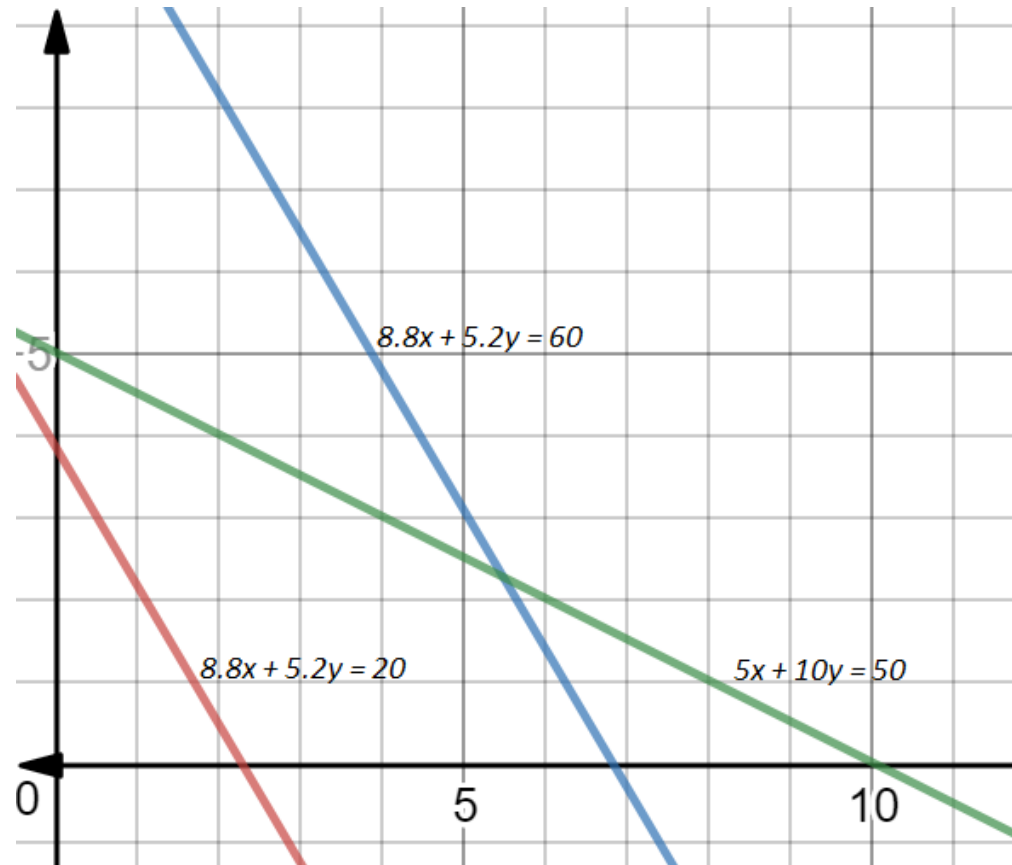


Graphical Method Example

- To check for the validity of the equations, put $x=0$, $y=0$ in (1).
- Clearly, it doesn't satisfy the inequality.
- Therefore, we must choose the side opposite to the origin as our valid region.
- Similarly, the side towards origin is the valid region for equation 2)
- **Feasible Region:** As per the analysis above, the feasible region for this problem would be the one in between the red and blue lines in the graph.

Graphical Method Example

- For the direction of the objective function; let us plot $5x + 10y = 50$.



Graphical Method Example

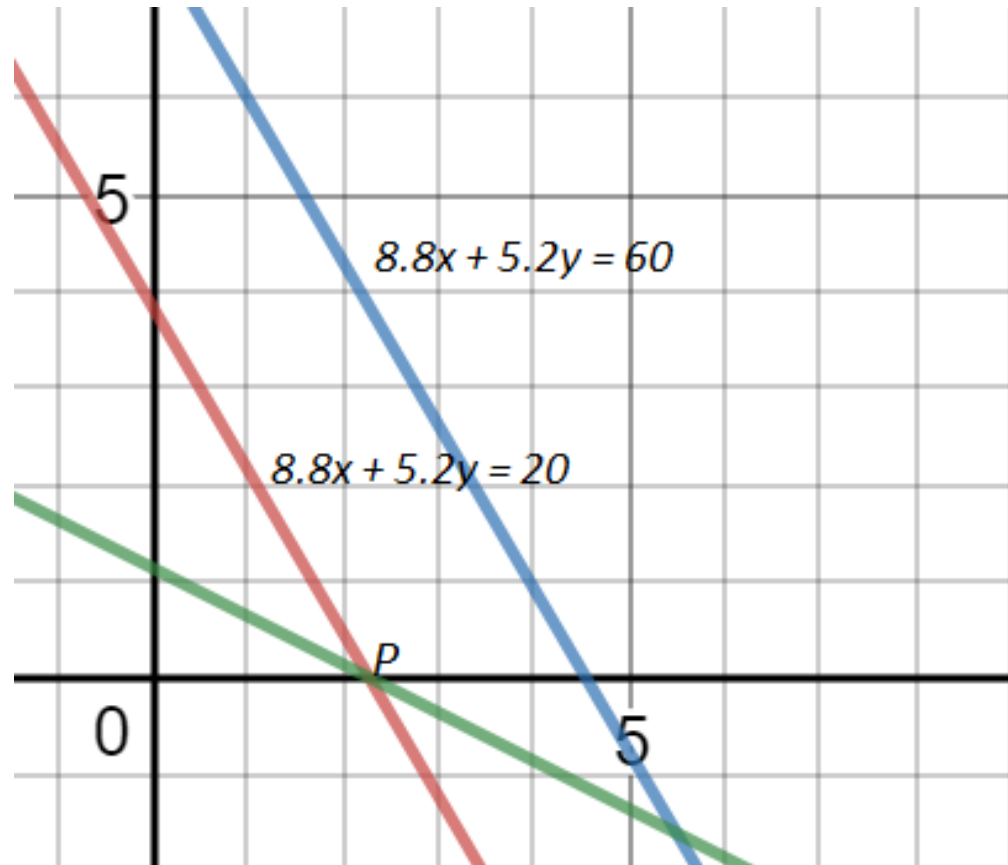
- Now take a ruler and place it on the straight line of the objective function.
- Start sliding it from the left end of the graph.
- We want the minimum value of the cost i.e. the minimum value of the optimum function C .
- Thus we should slide the ruler in such a way that a point is reached, which:
 - Lies in the feasible region
 - Is closer to the origin as compared to the other points

Graphical Method Example

- This would be our Optimum Point.
- It has been marked as P in the graph.
- It is the one which you will get at the extreme right side of the feasible region here.
- I've also shown the position in which your ruler needs to be to get this point by the line in green.

Graphical Method Example

- Check the graph.



Graphical Method Example

- Now coordinates of this point should be calculated.
- To do this, just solve the simultaneous pair of linear equations:

$$\begin{aligned}y &= 0 \\ 8.8x + 5.2y &= 20\end{aligned}$$

- We'll get the coordinates of 'P' as (2.27, 0).
- This implies that the family must consume 2.27 bananas and 0 apples to minimize their cost and function according to their diet plan.

Python Implementation

- **PuLP** is a python library which can be used to solve linear programming problems.
- Consider the following example,

$$\begin{array}{ll} \min & Z = 60x_1 + 40x_2 \\ \text{s.t.} & 4x_1 + 4x_2 \geq 10 \\ & 2x_1 + x_2 \geq 4 \\ & 6x_1 + 2x_2 \leq 12 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{array}$$